KAELEARN // ANALYSIS I // DEBRIEF

Arrows and Reasons

Eight problems, one habit: every arrow joins two statements and carries the reason that makes it true.



THE SESSION // READ THIS FIRST

- **Ex. 1**, $\mathbb{Z}_2$ is a field *(Körper)*: solved yourself; one case argued by a single instance.
- **Ex. 2**, $|xy| \leq \frac{x^2+y^2}{2}$ and $x^2 \leq y^2$: (a) clean; (b) right answer, but the key step has no reason.
- **Ex. 3**, three unit balls: sketches and $A_1 \subset A_2 \subset A_\infty$ right, read off the picture.
- **Ex. 4**, preimages *(Urbilder)*: (a) right idea, the arrows join the wrong things; (b) all eight right.
- **Ex. 5**, injective, surjective, bijective: all verdicts right; one inequality the wrong way round.
- **Ex. 6**, strictly monotone means invertible: right structure, wrong reason in one bracket.
- **Ex. 7**, $A \Longrightarrow B$ and solution sets: right answers; $B \Longrightarrow A$ refuted with half of A.
- **Ex. 8**, $\sum i^3$ by induction *(vollständige Induktion)*: correct; the step opens with the goal.
- **The pattern:** the mathematics is right almost everywhere. What slips is the grammar between the lines: $\Longrightarrow$ between numbers, $\Leftrightarrow$ between a statement and a set equation, a bracket that names the wrong hypothesis, the one step that *is* the exercise written without a reason. You found the cure yourself in Ex. 4: "NOT $\cup$, because this is a logical statement." Apply it to every arrow.
- **Again on 19 October:** one variant per problem on the last pages.
- The sheet was not in the Inbox. The problem statements below are reconstructed from your answers; check them against the sheet.

PROBLEM 01

Two Elements, Nine Axioms

THE PROBLEM // EX. 1 · RECONSTRUCTED

$\mathbb{Z}_2 = \{0, 1\}$ with $1+1 = 0$ and otherwise the usual rules: $0+0 = 0, 0+1 = 1+0 = 1$, $0 \cdot 0 = 0 \cdot 1 = 1 \cdot 0 = 0, 1 \cdot 1 = 1$. Show that $(\mathbb{Z}_2, +, \cdot)$ satisfies the field axioms.

Your path

You took the axioms one at a time and split each into cases: commutativity by $a \neq b$ and $a = b$, the neutral elements by trying both values, the inverses by listing them. For asso-

ciativity you found the right split: "some entry is 0" and "all are 1". In the multiplication version you argued the first case in general ($c = 0$, with a, b arbitrary). In the addition version you computed one triple, $(1, 0, 0)$, and wrote "analog".

WEAK POINTS // NAMED EXACTLY

01 **One instance standing in for a case.** "With 0" covers seven triples; $(1+0)+0 = 1 + (0 + 0)$ is one of them. *Fix:* inside a case, keep the other letters free. "Let $c = 0$. Then $(a + b) + 0 = a + b = a + (b + 0)$."

02 **One letter, two jobs.** In (Distr.) you set $a = 0$ and then wrote $0(a + b)$, with a now a summand. *Fix:* write the law with three letters, $a(b + c) = ab + ac$, then split on a.

THE DEEP IDEA // EXHAUSTION, SHORTENED BY THE NEUTRAL ELEMENTS

A finite structure can be checked case by case, and the neutral elements do most of the work: any case that contains a 0 collapses in one line, so only the all-ones case needs a computation. A case split is a proof when the cases cover everything *and* each case is argued for every one of its members.

Generalize it

- $\mathbb{Z}_p$ **for a prime** p is a field: addition and multiplication mod p. The one hard axiom is the multiplicative inverse, and it needs p prime. In $\mathbb{Z}_4$, $2 \cdot 2 = 0$, and 2 has no inverse: $2 \cdot 1 = 2$, $2 \cdot 2 = 0$, $2 \cdot 3 = 2$.
- $\mathbb{Z}_2$ **cannot be ordered.** In an ordered field *(angeordneter Körper)*, $1 = 1^2 > 0$, so $1 + 1 > 1 > 0$. In $\mathbb{Z}_2$, $1 + 1 = 0$. This is why $\mathbb{R}$ needs order axioms on top of the field axioms, and why Analysis starts with them.

Same structure, elsewhere

Case exhaustion is a truth table: Ex. 7 lives on one. Your three-case injectivity proof in Ex. 5b (both ≤ 0, both > 0, mixed) is the same move on infinite sets: the cases must cover every pair.

Why must $1 + 1$ *be* 0 *and not* 1*?*

Why?

If $1 + 1 = 1$, then 1 has no additive inverse: $1 + 0 = 1$ and $1 + 1 = 1$, so nothing added to 1 gives 0. Equivalently, $1 + 1 = 1 = 1 + 0$ and cancelling the first 1 would give $1 = 0$, which a field forbids.

Why?

Why is the case $a = b$ in commutativity "trivial"?

Because both sides are the same expression: $a + a = a + a$. Nothing about $\mathbb{Z}_2$ is used, only that $=$ is reflexive. That is also why the case $a \neq b$ is the only one that needs the table.

The solution, idea first

Idea: two elements, so every axiom is a finite check; the neutral elements dispose of every case with a 0.

Proof. *Neutral elements:* $0 + 0 = 0$, $1 + 0 = 1$, so $a + 0 = a$; $0 \cdot 1 = 0$, $1 \cdot 1 = 1$, so $a \cdot 1 = a$; and $0 \neq 1$. *Commutativity:* for $a = b$ both sides agree; for $a \neq b$ the table gives $1 + 0 = 1 = 0 + 1$ and $1 \cdot 0 = 0 = 0 \cdot 1$. *Inverses:* $0 + 0 = 0$ and $1 + 1 = 0$, so $-0 = 0$, $-1 = 1$; $1 \cdot 1 = 1$, so $1^{-1} = 1$. *Associativity of $+$:* if one of a, b, c is 0, both sides equal the sum of the other two, by $x + 0 = 0 + x = x$. Otherwise $a = b = c = 1$, and $(1 + 1) + 1 = 0 + 1 = 1 = 1 + 0 = 1 + (1 + 1)$. *Associativity of $\cdot$:* if one factor is 0, both sides are 0; otherwise both sides are 1. *Distributivity:* $a(b + c) = ab + ac$. For $a = 0$ both sides are 0; for $a = 1$ both sides are $b + c$. ∎

PROBLEM 02

Never Negative

THE PROBLEM // EX. 2 · RECONSTRUCTED ∎

Let $x, y \in \mathbb{R}$. (a) Show $|xy| \leq \frac{x^2+y^2}{2}$. (b) Show that $x^2 \leq y^2$ holds exactly when $|x| \leq |y|$, and describe it in the cases $y \geq 0$ and $y < 0$.

Your path

(a) is clean: you replaced x^2 by $|x|^2$, completed the square in $|x|$ and $|y|$, and dropped a square that is ≥ 0. In (b) your first instinct was $x \leq y$, and you struck it out in red yourself, which is the right reflex. Then the parabola picture, "closer or same distance to zero", which *is* $|x| \leq |y|$, and your note on the two methods for squares. The step $|x|^2 \leq |y|^2 \Rightarrow |x| \leq |y|$ is written down without a reason, and it is the whole of (b).

WEAK POINTS // NAMED EXACTLY ∎

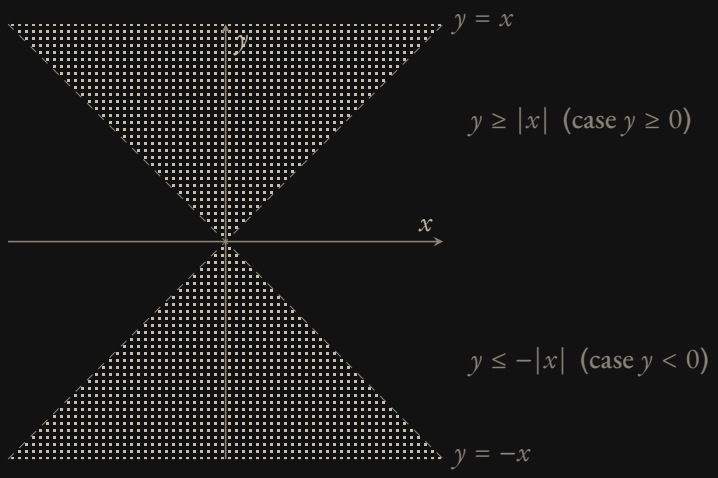
01 **The crux asserted.** $|x|^2 \leq |y|^2 \Rightarrow |x| \leq |y|$ is true, and it is what (b) asks you to show. *Fix:* when one step carries the problem, prove it as a lemma: for $u, v \geq 0$, $u^2 \leq v^2 \Leftrightarrow u \leq v$.

02 **Only one direction.** "Exactly when" asks for $\Leftrightarrow$; the page shows $\Rightarrow$. *Fix:* the lemma gives both at once.

THE DEEP IDEA // SQUARES ARE SAFE ON ONE SIDE OF ZERO

$t \mapsto t^2$ is strictly increasing on $[0, \infty)$, and a strictly increasing map preserves *and* reflects order (Problem 06). The absolute value moves x and y into $[0, \infty)$, where squaring is safe. For (a): to prove $A \le B$, write $B - A$ as a square:

$$\frac{x^2+y^2}{2} - |xy| = \frac{(|x|-|y|)^2}{2} \ge 0.$$



$x^2 \le y^2$ is the double wedge around the y-axis. Your two cases are its two halves. $x \le y$ would be everything above the dashed line $y = x$, a different set.

Generalize it

- **(a) is the mean inequality** *(AM-GM-Ungleichung)* for two numbers: $\sqrt{ab} \le \frac{a+b}{2}$ for $a, b \ge 0$ (set $a = x^2$, $b = y^2$). Equality exactly when $|x| = |y|$.
- **With a weight:** $|xy| \le \frac{1}{2}\left(\varepsilon x^2 + \frac{y^2}{\varepsilon}\right)$ for every $\varepsilon > 0$: apply (a) to $\sqrt{\varepsilon}\, x$ and $y/\sqrt{\varepsilon}$. Analysis uses this constantly, to shift weight between two terms.
- **Other powers:** $t \mapsto t^3$ is strictly increasing on all of $\mathbb{R}$, so $x^3 \le y^3 \iff x \le y$, with no absolute values. Even powers need $|\cdot|$; odd ones don't.

Same structure, elsewhere

Problem 03 uses $t^2 \le t$ on $[0, 1]$. Problem 05c needs $a^2 = b^2 \implies a = b$ for $a, b \ge 0$: the same lemma with $=$ in place of $\le$. Problem 06 proves the general fact behind it: strictly monotone maps reflect order.

Why does $x^2 \le y^2$ not give $x \le y$?

Squaring forgets the sign. $x = 1, y = -2$: $1 \le 4$, but $1 > -2$. Only the distance to 0 survives, which is $|x|$ and $|y|$.

Why?

Why do you write $|x|$ instead of x before completing the square in (a)?

Why?

The target has $|xy| = |x||y|$. In $(|x| - |y|)^2 = |x|^2 - 2|x||y| + |y|^2$ the cross term is exactly $2|xy|$. With $(x-y)^2$ you would get $2xy$, which proves $xy \le \frac{x^2+y^2}{2}$, a weaker statement when $xy < 0$.

The solution, idea first

Idea: (a) the difference is a square; (b) squaring is strictly increasing on $[0, \infty)$.

Proof. (a) $\frac{x^2+y^2}{2} - |xy| = \frac{|x|^2 - 2|x||y| + |y|^2}{2} = \frac{(|x|-|y|)^2}{2} \ge 0$.

(b) *Lemma.* For $u, v \ge 0$: $u \le v \Leftrightarrow u^2 \le v^2$. If $u \le v$, then $u^2 = u \cdot u \le u \cdot v \le v \cdot v = v^2$ (multiplying $u \le v$ by $u \ge 0$, then by $v \ge 0$). If $u > v$, then $u > 0$ and $u^2 = u \cdot u > u \cdot v \ge v \cdot v = v^2$; this is the contrapositive of "$\Leftarrow$".

Since $x^2 = |x|^2$ and $y^2 = |y|^2$, the lemma with $u = |x|$, $v = |y|$ gives $x^2 \le y^2 \Leftrightarrow |x| \le |y|$. If $y \ge 0$, this reads $|x| \le y$, that is $-y \le x \le y$. If $y < 0$, it reads $|x| \le -y$, that is $y \le x \le -y$. ∎

PROBLEM 03

Diamond, Disc, Square

THE PROBLEM // EX. 3 · RECONSTRUCTED ∎

Sketch $A_1 = \{(x,y) \in \mathbb{R}^2 : |x| + |y| \le 1\}$, $A_2 = \{(x,y) : |x|^2 + |y|^2 \le 1\}$ and $A_\infty = \{(x,y) : \max(|x|, |y|) \le 1\}$, and compare them.

Your path

You turned ≤ into = to get the boundary, then went quadrant by quadrant to get rid of the absolute values: four segments for A_1, two half-circles for A_2. For A_∞ you saw the key fact, "x and y are independent": $|x| \le 1$ and $|y| \le 1$ separately. Sketches right, $A_1 \subset A_2 \subset A_\infty$ right, and read off the picture.

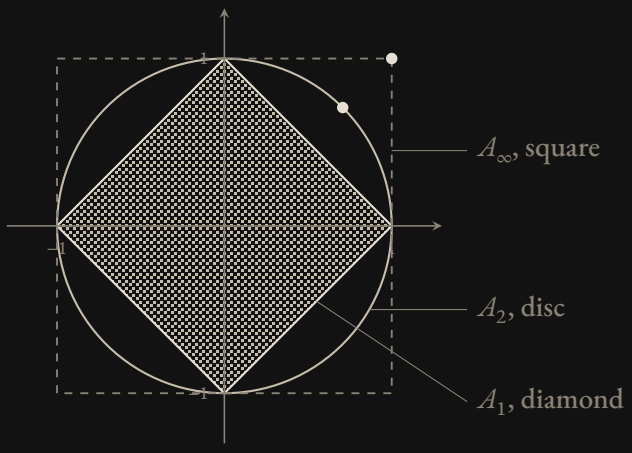
WEAK POINTS // NAMED EXACTLY ∎

01 **The inclusions are seen, not shown.** A sketch finds the claim; it does not prove it. *Fix:* for every ⊂ you draw, write the inequality that proves it. Here they are two lines.

02 ⊂ **hides a claim.** If you mean *proper* inclusion, name a point in the difference. *Fix:* $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \in A_2 \setminus A_1$ and $(1,1) \in A_\infty \setminus A_2$.

THE DEEP IDEA // ON $[0,1]$, SQUARING SHRINKS

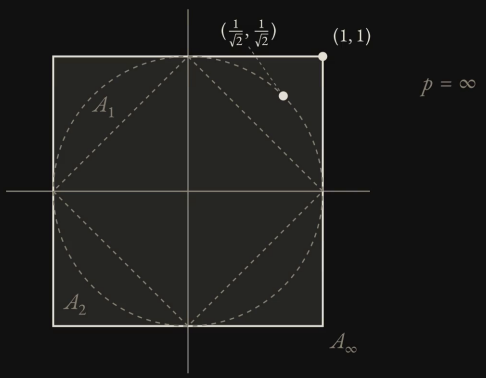
For $0 \le t \le 1$, $t^2 \le t$. So $|x| + |y| \le 1$ forces $x^2 + y^2 \le |x| + |y| \le 1$, and $x^2 + y^2 \le 1$ forces each $|x|, |y| \le 1$. These are the unit balls *(Einheitskugeln)* of three norms, $\|(x,y)\|_p = (|x|^p + |y|^p)^{1/p}$ and $\|(x,y)\|_\infty = \max(|x|, |y|)$. As p grows, the ball grows.



The diamond sits in the disc, the disc in the square. The two dots are the witnesses that the inclusions are strict: $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ *on the circle, outside the diamond;* $(1,1)$ *at the corner, outside the disc.*

ANIMATION // UNIT-BALLS · TAP TO PLAY

And $(1,1)$ is in the square, not the disc ($x^2 + y^2 = 2$).



Watch p run from 1 *to* ∞*: the diamond swells through the disc into the square, and never shrinks anywhere. 56 seconds.*

Generalize it

- **Every $p \geq 1$:** $A_p = \{|x|^p + |y|^p \leq 1\}$, and $p \leq q \Rightarrow A_p \subseteq A_q$. If $|x|^p + |y|^p \leq 1$, then $|x|, |y| \leq 1$, and for $t \in [0,1]$, $t^q \leq t^p$.
- **Below $p = 1$** the set is no longer convex (a four-pointed star), and $\|\cdot\|_p$ fails the triangle inequality: it is not a norm.
- **The other way round, after scaling:** $A_\infty \subseteq \sqrt{2}\, A_2 \subseteq 2A_1$. So the three norms agree on what "small" means, up to a constant. In $\mathbb{R}^n$ all norms do: norm equivalence *(Äquivalenz der Normen)*, and the reason a limit in the plane doesn't care which ball you use.

Same structure, elsewhere

Problem 02's lemma is the step $x^2 \leq 1 \Rightarrow |x| \leq 1$. Your boundary method (turn $\leq$ into $=$, then remove absolute values quadrant by quadrant) is case analysis again, as in Problem 01: four cases that cover the plane.

Why does the ball grow with p?

Why?

Points of the ball have $|x|, |y| \leq 1$, and on $[0,1]$ higher powers are smaller: $0.5^1 = 0.5$, $0.5^2 = 0.25$. Smaller left-hand side, easier to stay ≤ 1, more points qualify.

Why is A_∞ a square?

Why?

$\max(|x|, |y|) \leq 1$ holds exactly when $|x| \leq 1$ *and* $|y| \leq 1$: one condition per coordinate, independent of each other, as you wrote. Independent conditions on x and y give a product of intervals, $[-1,1] \times [-1,1]$.

The solution, idea first

Idea: on $[0,1]$, $t^2 \leq t$.

Proof. $A_1 \subseteq A_2$: let $|x| + |y| \leq 1$. Then $|x| \leq 1$, so $x^2 = |x| \cdot |x| \leq |x|$, and likewise $y^2 \leq |y|$. Hence $x^2 + y^2 \leq |x| + |y| \leq 1$. $A_2 \subseteq A_\infty$: let $x^2 + y^2 \leq 1$. Then $x^2 \leq 1$, so $|x| \leq 1$ by Problem 02's lemma; likewise $|y| \leq 1$. Both inclusions are proper: $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ has $x^2 + y^2 = 1$ but $|x| + |y| = \sqrt{2} > 1$, and $(1,1)$ has max $= 1$ but $x^2 + y^2 = 2 > 1$. ∎

PROBLEM 04

Preimages Follow Logic

THE PROBLEM // EX. 4 · RECONSTRUCTED ■

(a) Let $f : M \to N$ and $C, D \subseteq N$. Show $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$. (b) For $g : \mathbb{R} \to \mathbb{R}$, $g(x) = x^2$, find $g(\mathbb{R} \setminus \{0\})$, $g^{-1}([0,\infty))$, $g^{-1}(\{-2\})$, $g^{-1}(\{4\})$, $g^{-1}([-1,1])$, $g([-1,1])$, $g([0,1])$, $g^{-1}([0,1])$.

Your path

(a) A chain of equivalences through the definition, with your grey note at the middle step: ∨, "NOT ∪, because this is a logical statement". That distinction is the whole problem, and you saw it. (b) You began to write the image like a function, $g(\mathbb{R} \setminus \{0\}) : \mathbb{R} \setminus \{0\} \to \dots$, struck it out ("the image is just a set"), and then got all eight right. You wrote $g^{-1}([-1,1]) = [1]$ at first, and $\pm\sqrt{x}$ caught it.

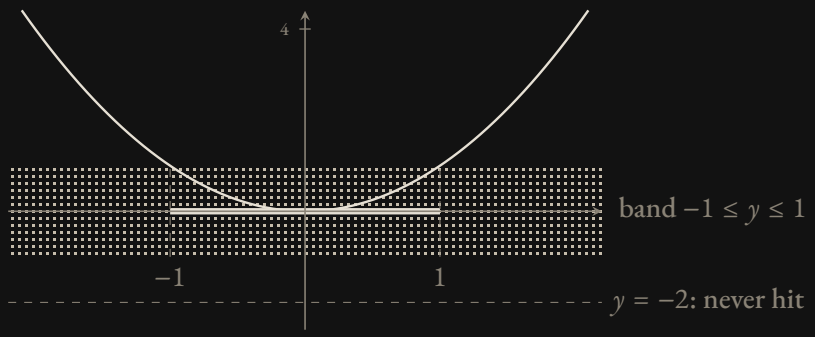
WEAK POINTS // NAMED EXACTLY

01 **"Let $x \in f^{-1}(C \cup D) \Leftrightarrow \dots$"** Once you assume x is in the left set, the backward direction runs on an assumption you no longer have. *Fix:* "Let $x \in M$. Then $x \in f^{-1}(C \cup D) \Leftrightarrow \dots$"

02 **The last ⇔ changes type.** It joins a statement about one x to an equation between sets. *Fix:* end the chain at $x \in f^{-1}(C) \cup f^{-1}(D)$. Then: "Since $x \in M$ was arbitrary, both sets have the same elements, so they are equal."

THE DEEP IDEA // A SET OPERATION IS A LOGICAL ONE IN DISGUISE

$f^{-1}(C)$ is defined by a condition on x: $f(x) \in C$. So every operation on C turns into an operation on that condition, ∪ into ∨, ∩ into ∧, complement into ¬, and preimages respect all of them. Images do not: $f(A) = \{f(x) : x \in A\}$ hides a "there is an x", and ∃ does not distribute over ∧.



$g^{-1}([-1,1])$ is every x whose height lands in the band: the thick segment $[-1,1]$. Only the top half of the band is ever reached, so $g([-1,1]) = [0,1]$. The line $y = -2$ is never hit: $g^{-1}(\{-2\}) = \emptyset$.

Generalize it

- **∩ and complement:** $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$ and $f^{-1}(N \setminus C) = M \setminus f^{-1}(C)$, by the same chain with ∧ and ¬.
- **Images:** $f(A \cup B) = f(A) \cup f(B)$ holds, but only $f(A \cap B) \subseteq f(A) \cap f(B)$. With g: $A = \{-1\}$, $B = \{1\}$, $g(A \cap B) = \emptyset$ but $g(A) \cap g(B) = \{1\}$. Equality for all A, B holds exactly when f is injective.
- **There and back:** $g^{-1}(g(A)) \supseteq A$, and can be bigger: $g^{-1}(g([0,1])) = [-1,1]$.

Same structure, elsewhere

Problems 01 and 07 also translate between sets and statements. Later in the course, continuity will be defined through preimages (preimages of open sets are open) for exactly this reason: preimages respect every set operation.

Why is $g^{-1}(\{4\}) = \{2, -2\}$ allowed when g has no inverse? Why?

$g^{-1}(C)$ is the name of a set, $\{x : g(x) \in C\}$, and it exists for every function and every C. It is not an inverse function; it only shares the symbol. When g is bijective the two meet: $g^{-1}(\{y\})$ is the one-point set containing the inverse's value.

Why $\vee$ and not $\cup$ in the middle of the chain? Why?

"$f(x) \in C$ or $f(x) \in D$" is a statement: it is true or false. $C \cup D$ is a set. $\cup$ joins sets, $\vee$ joins statements, and the definition $y \in C \cup D :\Leftrightarrow y \in C \vee y \in D$ is the bridge between them. Your note was exactly right.

The solution, idea first

Idea: unfold the definitions; $\cup$ becomes $\vee$.

Proof. (a) Let $x \in M$. Then $x \in f^{-1}(C \cup D) \Leftrightarrow f(x) \in C \cup D \Leftrightarrow f(x) \in C \vee f(x) \in D \Leftrightarrow x \in f^{-1}(C) \vee x \in f^{-1}(D) \Leftrightarrow x \in f^{-1}(C) \cup f^{-1}(D)$. Since x was arbitrary, the two sets are equal.

(b) $g(\mathbb{R}\setminus\{0\}) = (0, \infty)$: if $x \neq 0$ then $x^2 > 0$; and every $t > 0$ is $g(\sqrt{t})$ with $\sqrt{t} \neq 0$. Then $g^{-1}([0, \infty)) = \mathbb{R}$ since $x^2 \geq 0$ always; $g^{-1}(\{-2\}) = \emptyset$; $g^{-1}(\{4\}) = \{-2, 2\}$; $g^{-1}([-1, 1]) = \{x : x^2 \leq 1\} = [-1, 1]$; $g([-1, 1]) = [0, 1]$, since $x^2 \in [0, 1]$ there and $t = g(\sqrt{t})$ for $t \in [0, 1]$; $g([0, 1]) = [0, 1]$ by the same argument; $g^{-1}([0, 1]) = [-1, 1]$. ∎

PROBLEM 05

The Codomain Is Part of the Function

THE PROBLEM // EX. 5 · RECONSTRUCTED ∎

Decide whether each map is injective, surjective, bijective, and give the inverse where it exists. $f_1 : \mathbb{R} \to \mathbb{R}$, $f_1(x) = |x|$. $f_2 : \mathbb{R} \to \mathbb{R}$, $f_2(x) = x$ for $x \leq 0$ and $f_2(x) = \frac{1}{x}$ for $x > 0$. $f_3 : [0, \infty) \to [0, \infty)$, $f_3(x) = \frac{1}{1+x^2}$.

Your path

Every verdict is right. For f_1, $|-1| = |1|$ settles injectivity. For f_2 you solved $f(a) = b$ branch by branch, proved injectivity in three cases including the mixed one (the branches live on opposite sides of 0), and saw $f_2^{-1} = f_2$, with the grey warning that $f^{-1} \neq \frac{1}{f}$. For f_3 you solved $(1 + a^2)^{-1} = b$ for a and saw that b cannot exceed 1.

WEAK POINTS // NAMED EXACTLY

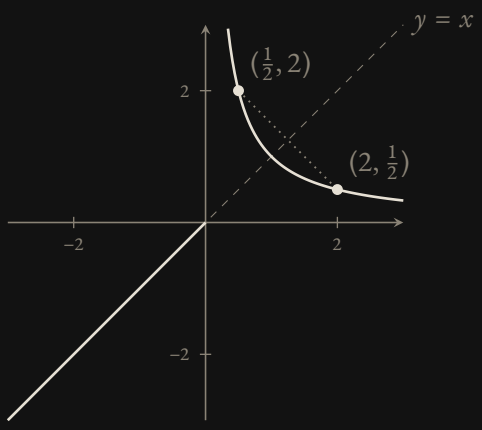
01 **5a: no named witness.** "If $b = -c \Rightarrow -c = |a|$" never says which b fails or why. *Fix:* one concrete value. $b = -1$: $|a| \geq 0 > -1$ for every a.

02 **5b: the $\exists$ is missing.** "$\forall b \leq 0$ s.t. $f(a) = b$" drops the quantifier that *is* surjectivity. *Fix:* write the pattern in full, $\forall b\, \exists a : f(a) = b$, and name the a.

03 **5c: reversed inequality.** "For a to be real, $\frac{1}{b} \leq 1$": the root needs $\frac{1}{b} - 1 \geq 0$, so $\frac{1}{b} \geq 1$, so $b \leq 1$. And $b = 0$ fails as well. *Fix:* when you solve for a, list every condition the formula needs ($b \neq 0$, radicand ≥ 0) and translate each into a condition on b.

THE DEEP IDEA // ONE EQUATION, TWO QUESTIONS

Surjective: $f(a) = b$ has a solution a *in the domain* for every b *in the codomain*. Injective: it never has two. Both are questions about one equation, and the answers depend on where a and b may live. Change the codomain and the verdict changes: f_3 is not onto $[0, \infty)$, but $f_3 : [0, \infty) \to (0, 1]$ is a bijection.



The graph of f_2 is its own mirror image in $y = x$: the ray lies on the mirror, and the branch of $\frac{1}{x}$ swaps $(\frac{1}{2}, 2)$ with $(2, \frac{1}{2})$. Mirroring a graph in $y = x$ draws the inverse, so $f_2^{-1} = f_2$.

Generalize it

- **Restrict to repair:** $|x|$ on $[0, \infty) \to [0, \infty)$ is the identity, bijective. $f_3 : [0, \infty) \to (0, 1]$ is bijective with inverse $b \mapsto \sqrt{\frac{1}{b} - 1}$.
- **Self-inverse maps** *(Involutionen)*: $f \circ f = \mathrm{id}$, graph symmetric in $y = x$. Others: $x \mapsto -x$, $x \mapsto c - x$, $x \mapsto \frac{1}{x}$ on $(0, \infty)$.
- **Finite versus infinite:** between finite sets of the same size, injective $\Leftrightarrow$ surjective. On $\mathbb{N}$ this fails: $n \mapsto 2n$ is injective and misses every odd number.

Same structure, elsewhere

Surjective means $f(M) = N$: an image computation as in Problem 04. Problem 06 gives the shortcut for 5c: f_3 is strictly decreasing on $[0, \infty)$, hence injective at once. Problem

02's lemma is the step $a^2 = b^2 \Rightarrow a = b$ for $a, b \geq 0$.

Why is f_2^{-1} not $\frac{1}{f_2}$?

Why?

f^{-1} undoes f: $f^{-1}(f(x)) = x$. $\frac{1}{f}$ is the reciprocal of the value. For f_2: $\frac{1}{f_2(-2)} = -\frac{1}{2}$, but $f_2^{-1}(-2) = -2$. Same symbol $^{-1}$, two different operations, as your grey note says.

Why does "not bijective" rule out an inverse?

Why?

An inverse $h : N \to M$ must satisfy $h(f(x)) = x$ and $f(h(y)) = y$. If $f(a) = f(b)$ with $a \neq b$, then h would have to send one value to both a and b. If some y has no preimage, $f(h(y)) = y$ is impossible. So an inverse needs both properties.

The solution, idea first

Idea: for each b, solve $f(a) = b$, then check that a is allowed.

Proof. f_1: $f_1(-1) = f_1(1)$ with $-1 \neq 1$, so not injective; $|a| \geq 0$, so -1 has no preimage and f_1 is not surjective. Not bijective, no inverse.

f_2: for $b \leq 0$ take $a = b$; then $a \leq 0$ and $f_2(a) = a = b$. For $b > 0$ take $a = \frac{1}{b} > 0$; then $f_2(a) = \frac{1}{a} = b$. So f_2 is surjective. On $(-\infty, 0]$, $f_2(x) = x$ is injective; on $(0, \infty)$, $\frac{1}{a} = \frac{1}{b} \Rightarrow a = b$; and a value from the first branch is ≤ 0, from the second > 0, so they never coincide. So f_2 is bijective. For $x \leq 0$, $f_2(f_2(x)) = f_2(x) = x$; for $x > 0$, $\frac{1}{x} > 0$ and $f_2(f_2(x)) = x$. Hence $f_2^{-1} = f_2$.

f_3: $f_3(a) = f_3(b) \Rightarrow 1 + a^2 = 1 + b^2 \Rightarrow a^2 = b^2 \Rightarrow a = b$, as $a, b \geq 0$. So injective. But $0 < f_3(a) \leq 1$ for all a, so $b = 2$ has no preimage: not surjective, not bijective, no inverse $[0, \infty) \to [0, \infty)$. ∎

PROBLEM 06

Order Reflected

THE PROBLEM // EX. 6 · RECONSTRUCTED ■

(a) Show that every strictly monotone $f : \mathbb{R} \to \mathbb{R}$ is injective. (b) Let $I, J \subseteq \mathbb{R}$ be intervals and $f : I \to J$ strictly monotone and surjective. Show that f has an inverse $f^{-1} : J \to I$ and that f^{-1} is strictly monotone in the same sense as f.

Your path

(a) Both cases, increasing and decreasing, straight from the definition, with the red "$x_1 \neq x_2$" added so it is clear what injectivity needs. (b) Bijective from (a) plus surjective, then a contradiction: suppose $x_1 \geq x_2$, push it through f, collide with $y_1 < y_2$. The structure is right. The bracket names the wrong hypothesis, and only the increasing case is done.

WEAK POINTS // NAMED EXACTLY

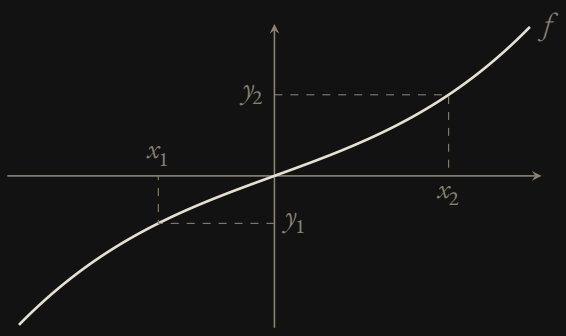
01 **"$f(x_1) \geq f(x_2)$ (since f is injective)".** Injectivity says $f(x_1) \neq f(x_2)$ for $x_1 \neq x_2$; it says nothing about which is larger. The step uses that f is *increasing*. *Fix:* after every step, name the hypothesis it uses, then check that the hypothesis really says that.

02 **Half the theorem, no conclusion.** Only increasing f is treated, and "so f^{-1} is strictly increasing" is never written. *Fix:* end with the sentence you proved; do the decreasing case by the same argument with the inequalities flipped.

03 **(a): from $x_1 \neq x_2$ to $x_1 < x_2$.** *Fix:* one phrase, "without loss of generality *(o.B.d.A.)* $x_1 < x_2$, otherwise swap the names".

THE DEEP IDEA // PRESERVING ORDER IS REFLECTING IT

A strictly increasing f preserves order, and therefore also reflects it: $f(x_1) < f(x_2) \implies x_1 < x_2$. The reason is that $\mathbb{R}$ is totally ordered: if not $x_1 < x_2$, then $x_1 \geq x_2$, and preservation gives $f(x_1) \geq f(x_2)$. Reflecting order is the same thing as "f^{-1} is increasing". Strict monotonicity is injectivity with a direction attached.



Read the graph backwards, from heights to positions: that is f^{-1}. Heights in order force positions in order, because if $x_1 \geq x_2$ an increasing f would give $y_1 \geq y_2$.

Generalize it

- **Drop "strictly":** a monotone function need not be injective. Constant functions are monotone.
- **Drop "surjective":** $f : I \to \mathbb{R}$ strictly monotone is still injective, so it is a bijection onto its image $f(I)$, and the inverse lives there.
- **The converse needs continuity.** Your f_2 from Problem 05 is injective and not monotone: it rises on $(-\infty, 0]$, falls on $(0, \infty)$, and jumps at 0. For a *continuous* injective function on an interval, strict monotonicity does follow, by the intermediate value theorem *(Zwischenwertsatz)*. Later in the course.

Same structure, elsewhere

Problem 02 is this theorem for $t \mapsto t^2$ on $[0, \infty)$: $u^2 \le v^2 \Leftrightarrow u \le v$ is reflection of order. Problem 05c's injectivity follows from f_3 being strictly decreasing. $\sqrt{}$, ln and arcsin will all be defined as inverses of strictly monotone functions, and this is why they are monotone too.

Why does preserving order give reflecting order? Why?

Any two reals compare: $x_1 < x_2$, $x_1 = x_2$ or $x_1 > x_2$. If $f(x_1) < f(x_2)$, the second option would give $f(x_1) = f(x_2)$, and the third $f(x_1) > f(x_2)$. Only the first is left. This needs a total order on the domain.

Why can't injectivity give $f(x_1) \ge f(x_2)$? Why?

Injectivity is about equality only. f_2 from Problem 05 is injective, and $-1 < 1$ gives $f_2(-1) < f_2(1)$, while $1 < 2$ gives $f_2(1) > f_2(2)$. Injective functions can do either.

The solution, idea first

Idea: strictly increasing maps reflect order; reflecting order is "f^{-1} increasing".

Proof. (a) Let $x_1 \neq x_2$, without loss of generality $x_1 < x_2$. If f is strictly increasing, $f(x_1) < f(x_2)$; if strictly decreasing, $f(x_1) > f(x_2)$. Either way $f(x_1) \neq f(x_2)$.

(b) By (a), f is injective, and it is surjective by hypothesis, so it is bijective and has an inverse $f^{-1} : J \to I$. Let f be strictly increasing, let $y_1 < y_2$ in J and $x_i = f^{-1}(y_i)$, so $f(x_i) = y_i$. Suppose $x_1 \ge x_2$. If $x_1 = x_2$, then $y_1 = y_2$. If $x_1 > x_2$, then $y_1 = f(x_1) > f(x_2) = y_2$, because f is increasing. Both contradict $y_1 < y_2$, so $x_1 < x_2$: f^{-1} is strictly increasing. For strictly decreasing f, the same argument with the inequalities flipped shows that f^{-1} is strictly decreasing. ∎

PROBLEM 07

Only Candidates

THE PROBLEM // EX. 7 · RECONSTRUCTED, CHECK AGAINST THE SHEET ■

For $(x, y) \in \mathbb{R}^2$ let A be the statement "$y = 1 - x$ and $y^2 = x^2 - 9$" and B the statement "$x = 5$". Show $A \Rightarrow B$. Does $B \Rightarrow A$ hold? Find the solution set of the system on $M_1 = \mathbb{R}^2$ and on $M_2 = \{(x, y) \in \mathbb{R}^2 : y \ge 0\}$.

Your path

$A \Rightarrow B$ by substitution: the squares cancel and $x = 5$. For $B \Rightarrow A$ you computed $y = 1 - 5 = -4$ and called it a contradiction with $y \ge 0$. Solution sets $L_1 = \{(5, -4)\}$ and $L_2 = \emptyset$, both right.

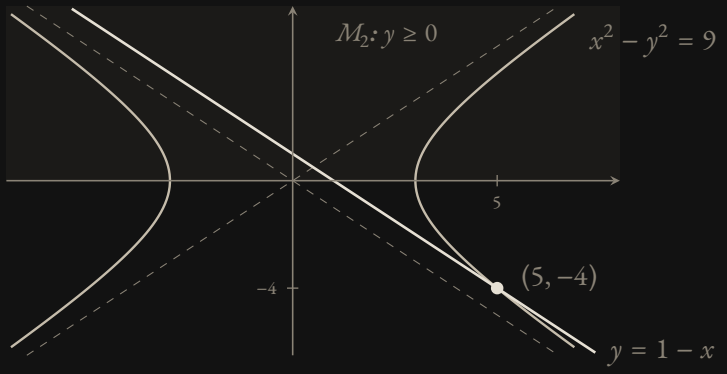
WEAK POINTS // NAMED EXACTLY

01 $B \Rightarrow A$ **refuted with A's own equation.** $y = 1 - x$ is half of A. From B alone, y can be anything. *Fix:* to refute an implication, give one point where the premise holds and the conclusion fails: $(5, 0)$. It works in M_2 too.

02 L_1 **from one direction only.** Solving showed "solution $\Rightarrow$ $(x, y) = (5, -4)$", that is $L_1 \subseteq \{(5, -4)\}$. That $(5, -4)$ really solves both equations is the converse, and it was never checked. *Fix:* plug candidates back in, or make every step $\Leftrightarrow$.

THE DEEP IDEA // SOLVING NARROWS DOWN, IT NEVER CONFIRMS

Solving an equation is a chain of implications from "(x, y) is a solution" to "(x, y) is one of these candidates". That is $A \Rightarrow B$: it narrows down. The converse, $B \Rightarrow A$, is what turns candidates into solutions, and steps such as squaring can break it. This exercise is that distinction in miniature.



The line meets the hyperbola once, at $(5, -4)$, below the shaded half-plane M_2. It is parallel to the dashed asymptote $y = -x$, which is why the x^2 cancelled and a single linear equation was left.

Generalize it

- **Squaring makes false candidates.** $\sqrt{x + 6} = x$: squaring gives $x^2 - x - 6 = 0$, so $x = 3$ or $x = -2$. Check: $\sqrt{9} = 3$ works, $\sqrt{4} = 2 \neq -2$ does not. The solution set is $\{3\}$.
- **Why only one point here:** a line parallel to an asymptote meets a hyperbola at most once. A line with any other slope can meet it twice, and then you would solve a quadratic.
- **The base set is part of the problem,** like the codomain in Problem 05: same equations, and on M_2 the solution set is empty. Formally $L_2 = L_1 \cap M_2$.

Same structure, elsewhere

Problem 05's surjectivity was the same discipline: solve $f(a) = b$, then check that the a is allowed. Problem 02 is the root cause of false candidates: squaring forgets the sign.

Why can't you use $y = 1 - x$ when testing $B \Rightarrow A$?

Why?

$B \Rightarrow A$ asks: does *every* point with $x = 5$ satisfy A? Those points are $(5, y)$ for every y. Using $y = 1 - x$ quietly restricts to points that already satisfy half of A, so you are testing a different implication, $B \wedge (y = 1 - x) \Rightarrow A$.

Why did the x^2 terms cancel?

Why?

$y^2 = x^2 - 9$ is the hyperbola $x^2 - y^2 = 9$, with asymptotes $y = \pm x$. The line $y = 1 - x$ has slope -1, parallel to $y = -x$. Substituting gives $x^2 - (1 - x)^2 = 9$, and the x^2 terms cancel exactly because the slopes match.

The solution, idea first

Idea: substitution gives the candidates; a counterexample refutes; plugging in confirms.

Proof. $A \Rightarrow B$: if $y = 1 - x$ and $y^2 = x^2 - 9$, then $(1 - x)^2 = x^2 - 9$, so $1 - 2x = -9$ and $x = 5$. $B \Rightarrow A$ is false: $(5, 0)$ satisfies B, but $0 \neq 1 - 5$. L_1: every solution has $x = 5$ and $y = 1 - x = -4$. Conversely $1 - 5 = -4$ and $(-4)^2 = 16 = 25 - 9$, so $L_1 = \{(5, -4)\}$. $L_2 = L_1 \cap M_2 = \emptyset$, since $-4 < 0$. ∎

PROBLEM 08

Factor What the Target Contains

THE PROBLEM // EX. 8 ∎

Show $\sum_{i=1}^{n} i^3 = \dfrac{n^2(n+1)^2}{4}$ for all $n \in \mathbb{N}$.

Your path

Base case, then the step: you split off $(n + 1)^3$, used the hypothesis, took a common denominator, and after one struck-out attempt found the move that makes it work. You factored out $(n + 1)^2$ (marked in red), and $n^2 + 4n + 4 = (n + 2)^2$ was left. Correct.

WEAK POINTS // NAMED EXACTLY ∎

01 **The step opens with the goal.** "$\sum_{i=1}^{n+1} \ldots = \frac{(n+1)^2(n+2)^2}{4} \Rightarrow \ldots$" writes what you want to prove and arrows away from it. Your computation never uses it, so the proof stands, but on an exam it reads as circular. *Fix:* write "To show: …" on its own line, then start from the left-hand side.

02 **$\Rightarrow$ between expressions.** $\frac{n^2(n+1)^2}{4} + (n + 1)^3 \Rightarrow \frac{\ldots}{4}$: these are numbers, not statements. *Fix:* a calculation is a chain of $=$.

03 **A dropped exponent:** the first sum reads $\sum i$ instead of $\sum i^3$.

THE DEEP IDEA // THE TARGET TELLS YOU WHAT TO FACTOR ■

In an induction step you know where you have to land. $\frac{(n+1)^2(n+2)^2}{4}$ contains $(n+1)^2$, so pull $(n+1)^2$ out of both terms at once; what is left must be $(n+2)^2$, and it is. Work from the left side to the target, with = between expressions.

Generalize it

- **Nicomachus:** $\sum_{i=1}^{n} i^3 = \left(\sum_{i=1}^{n} i\right)^2$, since $\sum i = \frac{n(n+1)}{2}$.
- **All powers:** $\sum_{i=1}^{n} i^k$ is a polynomial in n of degree $k+1$ with leading coefficient $\frac{1}{k+1}$: $\frac{n^2}{2}$, $\frac{n^3}{3}$, $\frac{n^4}{4}$, ..., like $\int_0^n x^k\,dx$.
- **Induction proves formulas; it doesn't find them.** To find one, compute small cases: 1, 9, 36, 100 are the squares of 1, 3, 6, 10.

Same structure, elsewhere

Problem 01 checked two elements one by one; induction is how a finite check reaches infinitely many. Problem 07 had the same direction rule: start from what is given, never from what is wanted.

Why may you use the formula for n inside the step? Why?

The step proves an implication, $P(n) \Rightarrow P(n+1)$, and $P(n)$ is its premise. Using $P(n+1)$, the conclusion, would be circular, which is why the step must not start from it.

Why is $\sum i^3$ a perfect square? Why?

With $T_n = \frac{n(n+1)}{2}$: $T_{n+1}^2 - T_n^2 = (T_{n+1} - T_n)(T_{n+1} + T_n) = (n+1)\cdot(n+1)^2 = (n+1)^3$, since $T_{n+1} + T_n = (n+1)^2$. So the squares T_n^2 grow by exactly the cubes, and both sides start at 1.

The solution, idea first

Idea: split off the last term, use the hypothesis, factor out $(n+1)^2$.

Proof. Let $P(n)$ be the statement $\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}$. $P(1)$: $1 = \frac{1\cdot 4}{4}$. Assume $P(n)$. Then

$$\sum_{i=1}^{n+1} i^3 = \sum_{i=1}^{n} i^3 + (n+1)^3 = \frac{n^2(n+1)^2}{4} + (n+1)^3$$
$$= \frac{(n+1)^2\left(n^2+4n+4\right)}{4} = \frac{(n+1)^2(n+2)^2}{4},$$

which is $P(n+1)$. By induction, $P(n)$ holds for every $n \geq 1$. ■

TRIGGERS // IF YOU SEE THIS, REACH FOR THAT

YOU SEE	REACH FOR
An inequality between squares	pass to $\lvert x\rvert, \lvert y\rvert$; $t \mapsto t^2$ is strictly increasing on $[0, \infty)$
Prove $A \leq B$	write $B - A$ as a square, or a sum of squares
Show two sets are equal	arbitrary x, chain $\Leftrightarrow$ on membership, then "x was arbitrary"
Is f surjective?	solve $f(a) = b$; list every condition the formula needs; check a is in the domain
You solved an equation by $\Rightarrow$ steps	only candidates so far: plug them back in
Refute an implication	one point where the premise holds and the conclusion fails
An induction step for a closed formula	factor out what the target contains

FOR THE DECK // MAKE THESE YOURSELF, THE MAKING IS THE LEARNING

TRIGGER	You see $x^2 \leq y^2$, or any inequality between squares.	Pass to $\lvert x\rvert, \lvert y\rvert$: $t \mapsto t^2$ is strictly increasing on $[0, \infty)$, so $x^2 \leq y^2 \Leftrightarrow \lvert x\rvert \leq \lvert y\rvert$.
WHY	Strictly increasing f: which property gives $x_1 \geq x_2 \Rightarrow f(x_1) \geq f(x_2)$?	Monotonicity. Injectivity only gives $\neq$, never an order.
TRIGGER	You solved an equation or a system by a chain of $\Rightarrow$.	You have candidates, not solutions. Plug them back in.

Again, in a week

Due 19 October 2026. No notes. Write on the grid, then bring it to your weekly review. The solutions are not in this file.

01 VARIANT OF EX. 1 In $\mathbb{Z}_3 = \{0, 1, 2\}$ with $+$ and $\cdot$ mod 3, give the multiplicative inverse of every nonzero element. In $\mathbb{Z}_4$, show that 2 has none. Which field axiom fails in $\mathbb{Z}_4$?

02 VARIANT OF EX. 2 Show $|x + y| \le |x| + |y|$ for all $x, y \in \mathbb{R}$ by comparing $(x + y)^2$ with $(|x| + |y|)^2$. Say where you use that both sides are ≥ 0.

03 VARIANT OF EX. 3 Sketch $B = \{(x, y) : |x| + 2|y| \le 2\}$ and $C = \{(x, y) : x^2 + 4y^2 \le 4\}$. Decide whether $B \subseteq C$ or $C \subseteq B$, and prove it.

04 VARIANT OF EX. 4 Let $f : M \to N$ and $C, D \subseteq N$. Show $f^{-1}(C \setminus D) = f^{-1}(C) \setminus f^{-1}(D)$. For $h : \mathbb{R} \to \mathbb{R}$, $h(x) = x^2 - 1$, find $h^{-1}([0, 3])$ and $h([-1, 2])$.

05 VARIANT OF EX. 5 Decide injective, surjective, bijective, and give the inverse where it exists: $k_1 : \mathbb{R} \setminus \{1\} \to \mathbb{R} \setminus \{1\}$, $k_1(x) = \frac{x+1}{x-1}$; and $k_2 : (0, \infty) \to (0, 1)$, $k_2(x) = \frac{1}{1+x^2}$.

06 VARIANT OF EX. 6 Let $f : I \to J$ be strictly decreasing and surjective. Show that $f^{-1} : J \to I$ exists and is strictly decreasing. After every step, name the hypothesis it uses.

07 VARIANT OF EX. 7 (a) For $(x, y) \in \mathbb{R}^2$ let A be "$x - y = 1$ and $x^2 - y^2 = 5$" and B be "$x = 3$". Show $A \Rightarrow B$. Does $B \Rightarrow A$ hold? Find the solution set on $\mathbb{R}^2$ and on $\{(x, y) : y < 0\}$. (b) Solve $\sqrt{x+2} = x$ over $\mathbb{R}$.

08 VARIANT OF EX. 8 Show by induction: $\displaystyle\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$ for all $n \in \mathbb{N}$.